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Experiment No 9

Aim

To create an ERC20 token in Remix IDE using the LearnEth Solidity ERC20 Token Course, fund a MetaMask wallet using a Sepolia testnet faucet, and deploy and interact with the token on the Sepolia testnet.

Theory

Token in the Blockchain Ecosystem

A token is a digital asset created and managed by a smart contract on an existing blockchain such as Ethereum, rather than by a blockchain of its own. The contract keeps a ledger of which address owns how many tokens and exposes functions to move them. Tokens can represent currency, voting rights, reputation, access rights or ownership of a real-world or digital item.

ERC20 Standard

ERC20 (Ethereum Request for Comments 20) is the standard for fungible tokens on Ethereum. Fungible tokens are all equal to each other and have the same value, behavior and rights, like the currencies ETH or BTC. Because every ERC20 token exposes the same interface, wallets, exchanges and other contracts can work with any of them without special integration. OpenZeppelin provides an audited, open-source implementation of this standard, which is imported and inherited in this experiment.

ERC20 Functions

- **totalSupply():** returns the total number of tokens in existence.
- **balanceOf(account):** returns the number of tokens owned by an account.
- **transfer(recipient, amount):** moves tokens from the caller to the recipient and returns a boolean indicating success.
- **approve(spender, amount):** allows a spender to withdraw up to a given amount from the caller's account.
- **allowance(owner, spender):** returns how many tokens the spender is still allowed to spend on behalf of the owner.
- **transferFrom(sender, recipient, amount):** moves tokens using the allowance mechanism; the caller must have been approved by the sender.

ERC20 Events

- **Transfer(from, to, value):** emitted whenever tokens move from one account to another, including minting (from the zero address) and burning (to the zero address).
- **Approval(owner, spender, value):** emitted whenever an allowance is set through approve().

ERC20 Attributes

- **name:** human-readable name of the token (EduCoin).
- **symbol:** short ticker of the token (EDC).

- **decimals:** number of decimal places used to display the token; the default is 18, so 1 EDC = 10^18 base units.

ERC20 vs ERC721

ERC20 tokens are fungible: every unit is identical and interchangeable, and ownership is tracked as a balance per address (for example currencies or reward points). ERC721 tokens are non-fungible: each token has a unique tokenId and distinct properties, and ownership is tracked per token through ownerOf(tokenId) (for example collectibles, game items such as baseball player cards, or deeds). ERC20 tokens can be divided into decimals, whereas an ERC721 token is indivisible.

Mintable, Burnable and Pausable Extensions

- **Mintable:** adds a mint function that creates new tokens and increases the total supply, normally restricted to an owner or minter role.
- **Burnable:** adds a burn function that permanently destroys tokens and reduces the total supply.
- **Pausable:** adds an emergency stop that lets an authorised account pause and unpause all token transfers.

Test Network and Wallet

MetaMask is a browser wallet that holds the account keys and signs transactions. Sepolia is an Ethereum test network whose ETH has no real value; it is obtained free from a faucet and pays the gas fee for deploying and using the token. Remix connects to MetaMask through the Injected Provider environment.

Program

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.3;

import "@openzeppelin/contracts@4.4.0/token/ERC20/ERC20.sol";

contract EduCoin is ERC20 {
    constructor() ERC20("EduCoin", "EDC") {
        _mint(msg.sender, 1000 * 10 ** decimals());
    }

    // Mintable: create new tokens
    function mint(address to, uint256 amount) public {
        _mint(to, amount);
    }

    // Burnable: destroy tokens held by the caller
    function burn(uint256 amount) public {
        _burn(msg.sender, amount);
    }
}
```

Outputs

Fig 1: Introduction chapter of the LearnEth Solidity ERC20 Token Course in Remix IDE, with the quiz questions answered correctly

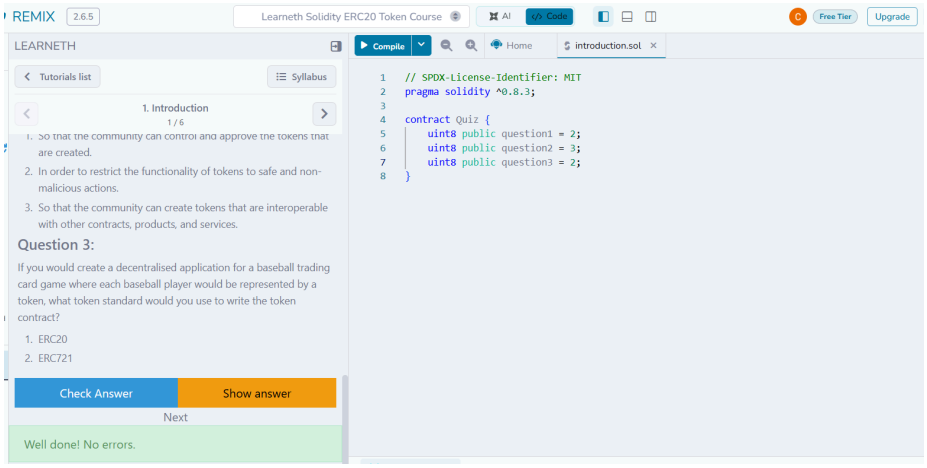


Fig 2: Interface chapter of the course, showing the IERC20 interface and its required functions

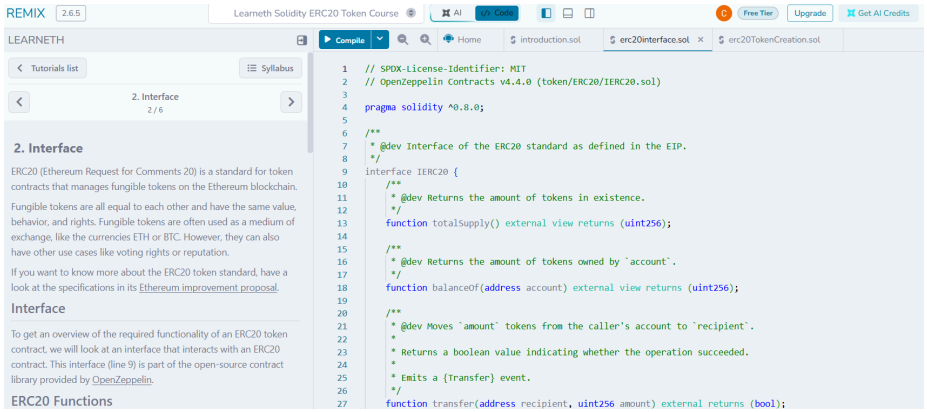


Fig 3: Token Creation chapter: EduCoin contract inheriting OpenZeppelin ERC20, with the assignment completed (token name "EduCoin", symbol "EDC", 1000 tokens minted)

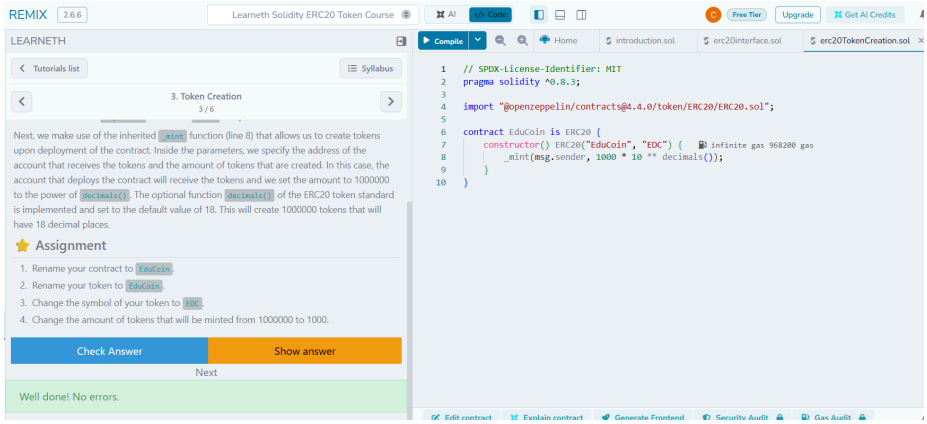


Fig 4: Compiling and deploying EduCoin on the Remix VM, with the transaction details (status, hash, block number and contract address) in the terminal

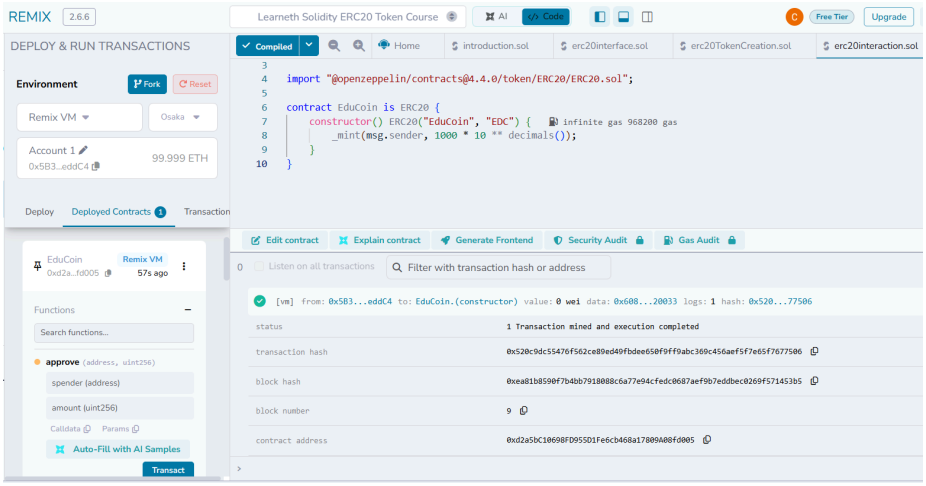


Fig 5: Calling name() on the deployed contract, which returns EduCoin

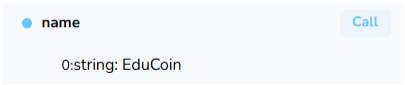


Fig 6: Calling symbol() on the deployed contract, which returns EDC

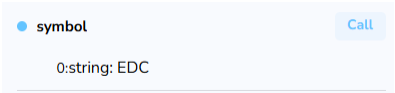


Fig 7: Calling totalSupply(), which returns 1000 × 10^18 base units (1000 EDC)

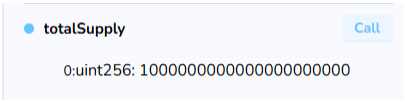


Fig 8: Calling decimals(), which returns 18

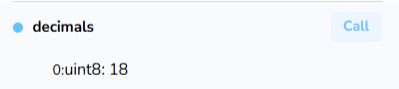


Fig 9: Calling balanceOf() for the deployer account, which holds the entire initial supply of 1000 EDC

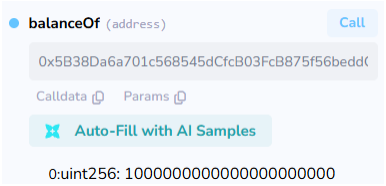


Fig 10: Deploying EduCoin to the Sepolia testnet from Remix (Injected Provider – MetaMask) and confirming the deployment in the MetaMask popup

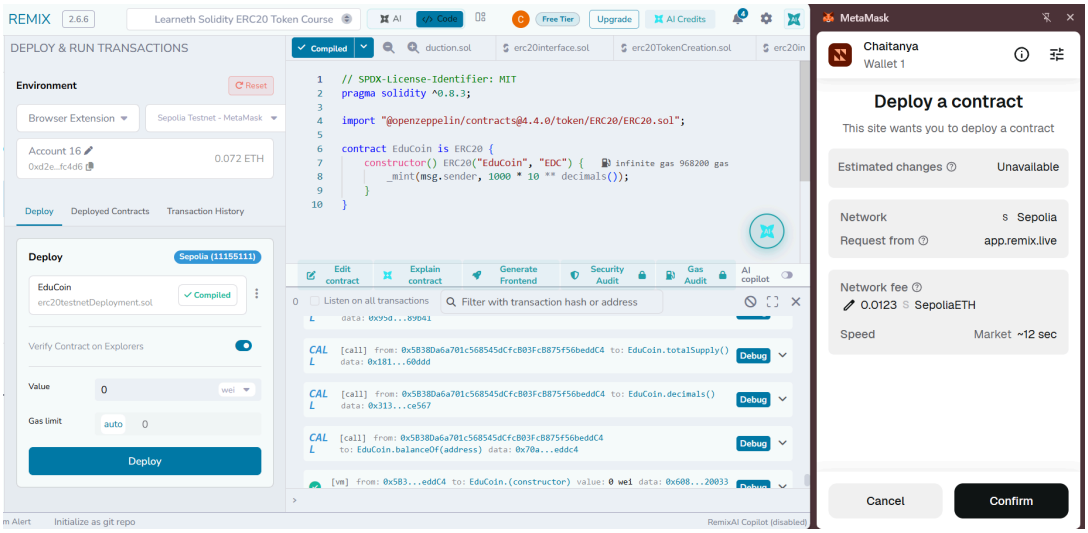


Fig 11: Adding the deployed EduCoin contract to MetaMask as a custom token (contract address, symbol EDC, 18 decimals)

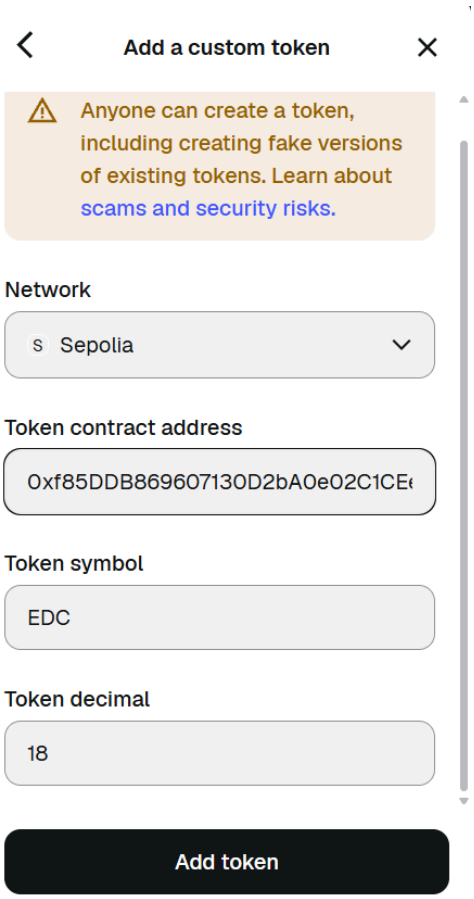


Fig 12: Calling transfer() from Remix and confirming the transaction in MetaMask to send EDC to another account (Account 3)

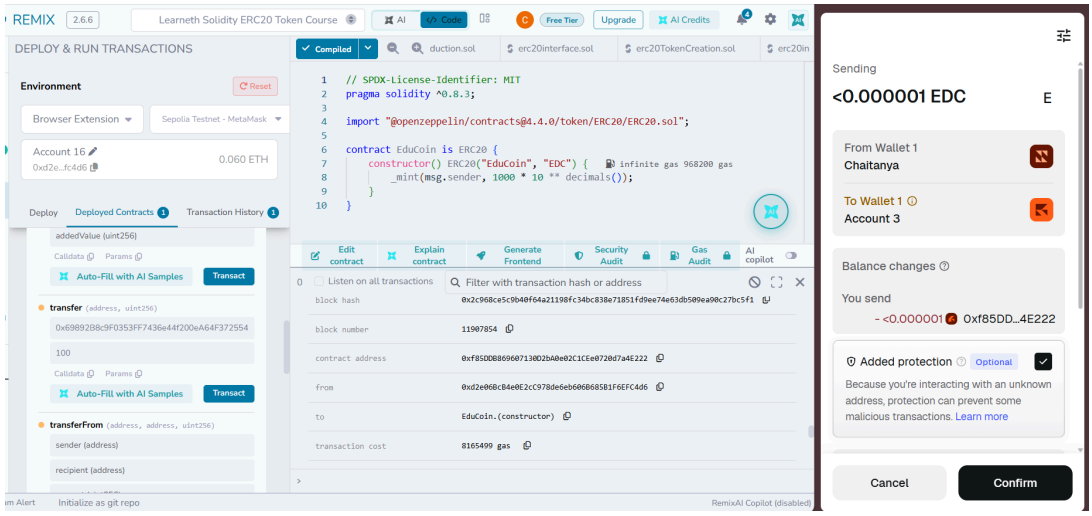


Fig 13: Transfer transaction mined on Sepolia; MetaMask Wallet 1 shows the EDC token balance

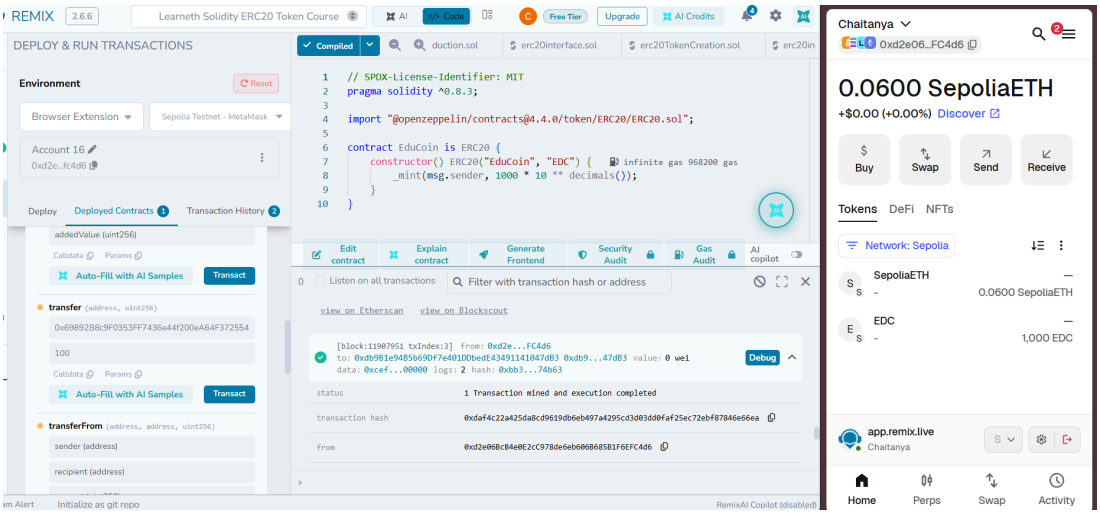


Fig 14: MetaMask view of Account 3 showing the EDC tokens received through the transfer

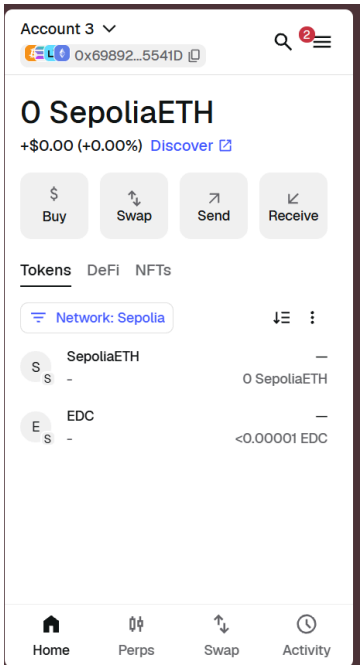


Fig 15: Calling mint() to create 100 new EDC tokens (100 × 10^18 base units)

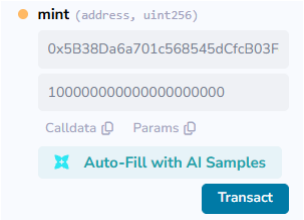


Fig 16: totalSupply() after minting, increased from 1000 EDC to 1100 EDC

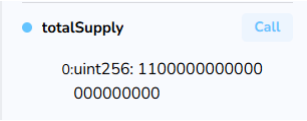


Fig 17: balanceOf() for the deployer account after minting, now 1100 EDC

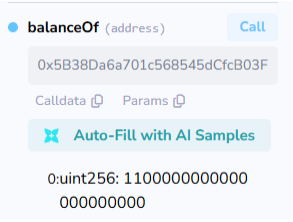
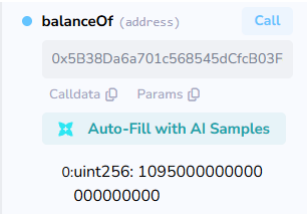


Fig 18: Calling burn() to destroy 5 EDC tokens and the successful burn transaction in the terminal



Fig 19: balanceOf() after burning, reduced to 1095 EDC



Conclusion

The ERC20 token EduCoin (EDC) was successfully created in Remix IDE by completing the six chapters of the LearnEth Solidity ERC20 Token Course, and was deployed to the Sepolia testnet using a funded MetaMask wallet. The token attributes (name, symbol, totalSupply and decimals) and balances were verified, tokens were transferred between accounts, 100 EDC were minted to raise the supply to 1100 EDC, and 5 EDC were burned to reduce the balance to 1095 EDC. The experiment also clarified the difference between the fungible ERC20 and non-fungible ERC721 token standards.